SESSION S04

Component Selection and the Mass Budget

Robotics, Drones & Autonomous Systems
KAUST Gifted Student Program · Summer Enrichment 2026

Tilt-rotor UAV · DARPA, public domain

WHERE WE ARE

The airframe is chosen.

Now determine whether it converges.

A QUESTION FOR THE ROOM

Can your aircraft carry its payload for the endurance you promised?

Answer if you are confident. **The answers are recorded and revisited at the design review** — not to catch anyone out, but because by then you will know.

60 SECONDS Answer now; the question returns at the design review

Session outline

- **Why sizing is a loop, and a worked example**
  Instructor-led. The only lecture in this session.
- **You size your aircraft, choosing components as you go**
  One exercise, not two. The components are inputs to the loop.
- **Submit Report 2**
  A hard stop. Work is collected as it stands.
- **Design review**

Output: Report 2 — the components you chose and the mass budget that proves they close. With S03's airframe decision it is worth **20%**. There is no homework in this course, so what is on the table at the bell is what is marked.

PART ONE

Why sizing requires iteration

The battery appears on both sides of the equation

To find your takeoff mass you need the battery mass. To find the battery mass you need the power. To find the power **you need the takeoff mass.**

Heavier aircraft	→	more power to stay up
More power	→	more energy for the same endurance
More energy	→	a bigger battery → *heavier aircraft*

> That is a **loop**, not a sum — and in S03 you chose a rotor count and a propeller size. **This is where you find out what they cost you.**

The sizing loop

01

Guess a takeoff mass

Any sensible number. The loop does not care how good the guess is, only that you are honest about what comes back.

02

Pick a powertrain that can lift it

Use **your own thrust-to-weight target** and **your rotor count**, both from S03, to get the thrust each motor must produce. Choose motor and propeller from the catalogue — this is where components enter.

03

Work out the power to stay up

Hover power for a rotorcraft, cruise power for a wing. Then add your payload's power and the avionics.

04

Energy, then battery mass

Power × endurance = energy. Energy ÷ usable specific energy = the mass of battery you must carry.

05

Add it all up, then compare

Structure + propulsion + battery + payload + avionics. **Does it match your guess?** If not, that answer is your next guess.

Hover power, including losses

HOVER POWER

$$P = \frac{(mg)^{3/2}}{\sqrt{2\rho A}\ \cdot\ \mathrm{FM} \cdot \eta}$$

P electrical power, W · **m** takeoff mass, kg · **g** 9.81 m/s^2 · **ρ** air density, 1.225 kg/m^3 · **A** total disc area, m^2 · **FM** figure of merit, 0.65 · **η** motor + ESC efficiency, 0.85

A is total disc area — every rotor added together, which is where **S03's rotor count and propeller diameter** arrive.

WHAT IT IS TELLING YOU

Power does not rise in proportion to weight. It rises with **weight to the power 1.5** — add 20% mass and you pay about **33% more power**.

And it falls with **√A**. That is S03's disc loading, and it is why the propeller you chose is the largest single lever you have here.

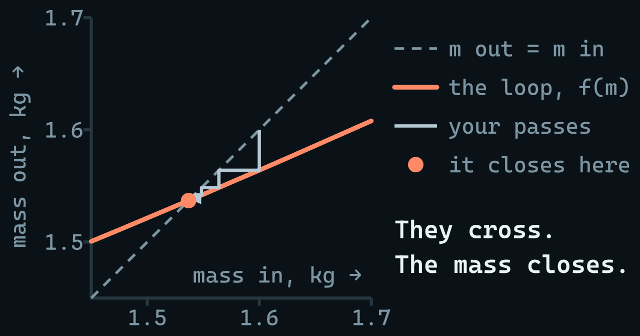
An **X8** takes its interference loss off the top — your disc area is not eight propellers' worth.

Every constant here is on your component card. **Nothing on this slide is something you have to look up** — the work is deciding what to put into it.

A converging case

Worked example — **not your brief**. A quadcopter on four 10-inch propellers, carrying a 300 g sensor drawing 6 W, target endurance **25 minutes**, thrust-to-weight 2:1 — so each of the four motors must deliver about **780 g**.

PASS	MASS GUESSED	POWER	ENERGY	BATTERY	MASS OUT
1	1.60 kg	171 W	71.1 Wh	0.494 kg	1.56 kg
2	1.56 kg	165 W	68.9 Wh	0.478 kg	1.55 kg
	Converged				**1.55 kg**



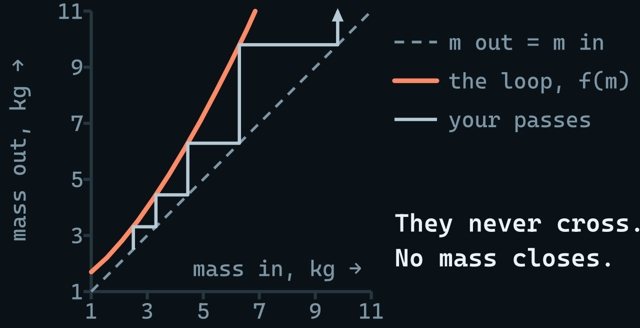
Two passes. The mass that came out matched the mass that went in to within one per cent, and **that is what "it closes" means.** You now have a real aircraft: a takeoff weight, a powertrain, and a battery you could go and buy.

The picture is the same two passes. Plot what comes **out** against what went **in**, and the loop closes wherever that curve meets the line where the two are equal. **Every pass walks you onto it.**

A diverging case

The **same aircraft**. Same propellers, same sensor, same battery chemistry. One requirement changed: endurance is now **60 minutes**.

PASS	MASS GUESSED	POWER	ENERGY	BATTERY	MASS OUT
1	2.50 kg	323 W	323 Wh	2.24 kg	3.31 kg
2	3.31 kg	487 W	487 Wh	3.38 kg	4.45 kg
3	4.45 kg	752 W	752 Wh	5.22 kg	6.29 kg
4	**6.29 kg**	**1257 W**	**1257 Wh**	**8.73 kg**	**9.80 kg**



They never cross.
No mass closes.

> Nothing is wrong with the arithmetic. **The aircraft does not exist.** Every extra gram of battery costs more power than the energy it carries buys back — and no number of passes will ever bring it back.

The same plot as the last slide — read the numbers on the axes. **The curve never touches the line**, so no mass anywhere returns itself, and the passes are climbing a staircase with no top.

Three legitimate responses to divergence

01

Grow the disc

Power falls with $\sqrt{A}$. Bigger propellers, or more of them, cost you mass and folded size — but they attack the problem at its source. This is the answer the physics likes.

02

Change the chemistry

Li-ion carries roughly **50% more energy per kilogram** than LiPo. It will not give you the burst current a heavy quadcopter wants on takeoff, so you are trading margin for endurance. Say which you chose.

03

Go back to the requirement

Sixty minutes may simply not be purchasable in this configuration at this budget. **Name the requirement, say what it costs, and propose the number that does close.**

> Option 03 is **not** giving up. A team that shows exactly what an extra ten minutes of endurance costs has done engineering. A team that quietly writes down a smaller battery has not.

PART TWO

Exercise — 40 minutes

The component catalogue

An extract. The full card on your table carries propellers, airframes and every constant.

Motors — thrust at the stated propeller

ID	PROP	THRUST	POWER	MASS
M1	10×4.5	850 g	145 W	55 g
M2	11×4.7	1050 g	190 W	68 g
M3	13×4.4	1900 g	320 W	105 g
M4	15×5.5	2600 g	430 W	165 g

Batteries — usable energy is 80% of stated

ID	TYPE	ENERGY	MASS	WH/KG
B1	LiPo 4S	74 Wh	480 g	154
B2	LiPo 6S	111 Wh	700 g	159
B3	Li-ion 6S	151 Wh	640 g	236
B4	Li-ion 4S	151 Wh	660 g	229

Four motors on a quadcopter means **four times the mass and four times the power** — and the thrust column is *per motor*. Read the units twice.

The mass budget template

Sheet 6 — The sizing loop

One row per pass. Keep every pass.

PASS	1, 2, 3 ...
MASS IN	Your guess for this pass
POWER	From the equation, in watts
BATTERY	Energy ÷ usable Wh/kg
MASS OUT	The sum. Next pass's guess

Sheet 7 — Mass budget & components

What the aircraft is made of.

ITEM	Structure · propulsion · battery · payload · avionics
CHOICE	Catalogue ID, where there is one
MASS	Grams, and they must sum
WHY	One line. The line is the mark

Keep the passes that did not work. A sheet showing one pass is a guess written down neatly. The convergence — or the runaway — is the evidence you did this at all.

Task

CORE · EVERYONE FINISHES THIS

STRETCH · IF YOU FINISH EARLY

Task

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- Take your payload **mass and power** and endurance target from your brief, and your **rotor count and thrust-to-weight** from S03

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- Run the loop. Record **every pass**, until it converges or clearly does not

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- Write down the endurance your sizing supports, **next to the one you promised in S02**

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- Change one requirement by 20% and re-run. Which one is your design most sensitive to? **Name it.**

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STRETCH · IF YOU FINISH EARLY

- Change one requirement by 20% and re-run. Which one is your design most sensitive to? **Name it.**
- Does the airframe you chose in S03 survive its own sizing? If not, say what you would change — the rotor count, the propeller, or a number in the brief

PART THREE

Submission and review

Design review — four minutes per team

Tell the room:

1. Your takeoff mass, and the heaviest single line in your budget
2. The endurance your sizing supports, against the one your brief demands
3. Whether S03's airframe survived — and if not, what you changed
4. The number you are least confident in

Question 4 again. In S01 it was *what would change your mind*; in S03 it was *what evidence would change your mind*; here it is **which of your own numbers do you distrust.** Four sessions, one habit.

THAT IS THE PAPER DESIGN

Four sessions, two projects

In S01–S02 a problem statement became a **class of aircraft**. In S03–S04 that class became a **particular machine** — a rotor count, a propeller, a battery and a mass — and the mass told you whether the mission was ever possible.

> Nothing in either report was a lookup.
> **Every line of both was a decision you can defend.**

Next the paper stops and the hardware starts.

Photographs

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